

# WILL MCGOWAN

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## EDUCATION

University of Strathclyde, Glasgow

Sep 2023 – Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1, including 90% in first-year Finance
- **Relevant Modules:** Applied Econometrics, Treasury Management and Derivatives, Advanced Microeconomics

**Dissertation** | MATLAB

In progress, due 2027

*"Who Hedges the Barrel? Sovereign Oil Hedging and Petro-Currency Reactions to OPEC News"*

- Testing whether sovereign hedges reshape a petro-currency's oil beta: smaller under Norway's krone conversions (vs unhedged Canada), kinked at Russia's budget-rule cut-off, one-sided under Mexico's puts
- Measuring each currency's move on OPEC announcement days since 1983, with difference-in-differences around rule changes and the dollar's own move stripped out
- Testing a pre-registered trading rule on any residual beta after costs, with a White reality check and Kelly sizing

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

## PROJECTS

Plains Heat and the KC-Chicago Wheat Spread

R

- Tested whether Plains heatwaves move Kansas City wheat futures against Chicago, using 27 summers of NOAA weather data and the off-season as a placebo: heat explained 0.3% of weekly moves
- Found a one-day timestamp error in the price data that would have faked a lead-lag signal; with it fixed, a trading rule set in advance showed no edge out of sample (2010–23)

Delta-Hedging Simulator

MATLAB

- Simulated delta-hedging a short call on 50,000 paths: hedging four times as often halved the hedging error, and 0.1–0.5% trading costs moved the best frequency from daily to weekly
- Tested a no-trade band around delta instead of a fixed timetable: on fresh paths the best band cut RMS P&L by 6% vs the best timetable at 0.1% costs, and by 2% at 0.5%

Monte Carlo Option Pricing Engine

MATLAB

- Priced an arithmetic Asian call by Monte Carlo, validated against two closed-form prices; a geometric-Asian control variate cut the variance 1,276-fold (97% off the standard error)
- Mapped where the control variate weakens over 8 volatilities and 6 strikes: at the money its gain falls from 4,569x to 71x as volatility rises from 10% to 80%, and to 52x at worst out of the money

Empirical CAPM Test

MATLAB

- Tested the CAPM on 694 anomaly portfolios in 54 countries (JKP data): alpha significant in 31% at 5%, over six times the rate expected by chance; GRS rejected in 48 of 53 countries
- Corrected for running 694 tests at once: 123 of the 215 significant alphas survive Benjamini-Hochberg and 84 clear  $|t| > 3$ ; momentum holds up best, while size's hits are no more than luck predicts

## WORK EXPERIENCE

Le Set, Glasgow | Head Mixologist

Sep 2026 – Present

- Wrote a new 200-cover venue's 11-drink menu, including 4 original cocktails; trained around 15 bar staff

KONG, Glasgow | Mixologist and Front of House

Oct 2024 – Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%

## LEADERSHIP AND ACTIVITIES

Strathclyde Agricapital Society | Co-Founder and Treasurer

Sep 2025 – Present

- Co-founded Strathclyde's first agri-finance society, grew it to 126 members and published weekly research notes

Strathclyde Poker Society | Vice-President, then President

Sep 2025 – Apr 2026

- Ran weekly tournaments for around 60 players on £600 a semester, reconciling every buy-in and prize pool

## ADDITIONAL INFORMATION

**Technical:** R, MATLAB, Excel, Git

**Chess:** 14th at the 2023 Irish U19 Championships; played 22 opponents at once (average 1860 FIDE) and won 21

**Poker:** multi-tabled 12 cash games at once; used the Kelly criterion to set stakes from win rate and variance, but stayed below full Kelly (a bankroll of at least 50 buy-ins) in case the edge was overestimated

**Powerlifting:** Northern Ireland U19 champion (2023)